

The empirical recurrence for OEIS A250971, recovered from its tail

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Abstract

OEIS A250971 records “Empirical recurrence of order 57” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 128$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 59$. Its last coefficient is $c_{57} = 898560000 \neq 0$, so no further tail satisfies anything shorter; any order-57 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A250971 is “Number of $(n+1) \times (6+1)$ 0..1 arrays with no 2×2 subblock having the sum of its diagonal elements less than the absolute difference of its antidiagonal elements.”. It has offset 1 and begins

7448, 489543, 32124018, 2112130109, 138900769559, 9135168475316, 600808386292820, 39514486757351514, ...

The entry states:

Empirical recurrence of order 57 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:55:37, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 128$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 128$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 57: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 57.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 256$ exact terms from $n = 2$ on, it returns order 57 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{57} c_i a(n-i),$$

c_1	128	c_{16}	−71574584599657660	c_{31}	1938054815061254258215	c_{46}	−1270141137
c_2	−6284	c_{17}	34919228714343072	c_{32}	−2210463277386217790937	c_{47}	3512865150
c_3	202384	c_{18}	426604584194045621	c_{33}	2463792872630231771759	c_{48}	−926441844
c_4	−5056406	c_{19}	−2206162199945167137	c_{34}	−2525009278406567439648	c_{49}	2256717778
c_5	101277261	c_{20}	6633989254001729072	c_{35}	2299444058517643525245	c_{50}	−47691748
c_6	−1660926065	c_{21}	−13743547010315423125	c_{36}	−1851100752223304037377	c_{51}	864585228
c_7	22719653175	c_{22}	17301295868120720843	c_{37}	1335934516420852166120	c_{52}	−15060492
c_8	−260242351031	c_{23}	2875082712040073334	c_{38}	−883139347931241594519	c_{53}	27868946
c_9	2493010819801	c_{24}	−84800594894160467375	c_{39}	543533624844930109575	c_{54}	−4614862
c_{10}	−19959532958437	c_{25}	272078454026419988849	c_{40}	−312466760053384550209	c_{55}	5271893
c_{11}	133261575374623	c_{26}	−579882349018992032932	c_{41}	166209480679559045898	c_{56}	−338670
c_{12}	−738612290471348	c_{27}	960636841643917831414	c_{42}	−80315188295112378806	c_{57}	898560
c_{13}	3374242263588094	c_{28}	−1314929228920281440989	c_{43}	34509029677335651319		
c_{14}	−12512488819869047	c_{29}	1566609936468764087821	c_{44}	−12987126908575776820		
c_{15}	36168118379088315	c_{30}	−1740277345035624901778	c_{45}	4286317945022680269		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 59$, and it is the recurrence OEIS A250971 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{57} - c_1 x^{56} - \dots - c_{57}$. Its constant term is $-c_{57} = -898560000$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 57 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 57, so $\deg q = 57$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 57, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 12 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 57, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 898560000.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-57 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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